CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

B.Tech. (CL/EC/ME/CE/EE/IT) Sem - II Examination, May-2012

ME - 101 Engineering Graphics

Date: 21/05/2012, Monday **Time: 10.00 a.m to1.00 p.m.** **Maximum Marks: 70**

Instructions: (i) Attempt all the questions.
(ii) Make suitable assumptions if require

SECTION 1

Q.1 Choose appropriate answer by rewriting the question. **[5]**

1 Hidden lines are represented by ______ lines
(a) Thin Dotted (b) Center (c)Chain (d)Cutting

2 When a cone is cut by a section plane that is parallel to the axis and passing through it, their intersection curve is ------
(a)Cycloid (b)isosceles triangle (c)Spiral (d)Involute

3 In first angle projection, the object is placed in between the observer and the plane of projection (True/false)

4 A prism is named according to the shape of its ends (True/False)

5 The proportions of the arrow head of the dimension line is _____
(a) 3:1 (b)4:1 (c) 5:1 10:1

OR

Q.1 i) Construct a scale of R.F. = 1:60 to read Yards and feet, and long enough to measure up to 5 yards. Indicate 3 yard and 1 foot on it. [3]

ii) Give the application of AutoCAD [2]

Q. 2 (a) The major axis of an ellipse is 150 mm long and the minor axis is 100 mm long. Draw the ellipse by concentric circle Method. [8]

Q. 2(b) A line PQ 60 mm long has its end P in the V.P. and the end Q in the H.P. The line is inclined at 30^0 to the H.P. and at 60^0 to the V.P. Draw its Projection. [10]

OR

Q. 2(b) A line AB, 90 mm long, is in the H.P. and makes an angle of 30^0 with the V.P. its end A is 25 mm in front of the V.P. Draw its Projections. [10]

Q. 3 Fig. 1 shows pictorial view of an object. Draw [12]
(1) Front view in the direction 'X', (2) Top view.

SECTION 2

Q.4 A rectangular plate of sides 60mm*30mm has its shorter side in the H.P. and inclined at 30^0 to the V.P. Project its front view if its top view is a square. [07]

OR

Q.4 A regular pentagonal plate of 100 mm sides has one of its corner on HP the plane of the pentagon is inclined at 30° to the HP the side of the pentagon which is opposite to the corner which is on HP is inclined at 45° to the VP. Draw the projection of plate. [07]

Q.5(a) A pentagonal pyramid, side of base 50 mm and height 80 mm rests on its base on the ground with one of its base sides parallel to V.P. A section plane perpendicular to VP and inclined at 30° to H.P cuts the pyramid, bisecting its axis. Draw the development of the pyramid and sectional top view. **[09]**

(b) A square prism with side of base 30 mm and axis 50 mm long has its axis inclined at 60^0 to H.P. on one of the edges of the base which is inclined at 45^0 to V.P. **[09]**

OR

Q.5(a) A pentagonal prism of side of base 25mm and axis 40mm long is resting on HP on a corner of its base. Draw the projections of the prism, when the base is inclined at 60° to HP., and the axis appears to be inclined at 30° to V.P. **[09]**

(b) A hexagonal pyramid with side of base 30 mm and height 75 mm stands with its base on H.P. and an edge of the base parallel to V.P. It is cut by a plane perpendicular to V.P. inclined at 45° to H.P and passing through the mid-point of the axis. Draw the sectional top view and develop the lateral surface of the pyramid. **[09]**

Q.6 Draw the isometric view for fig-2. **[10]**

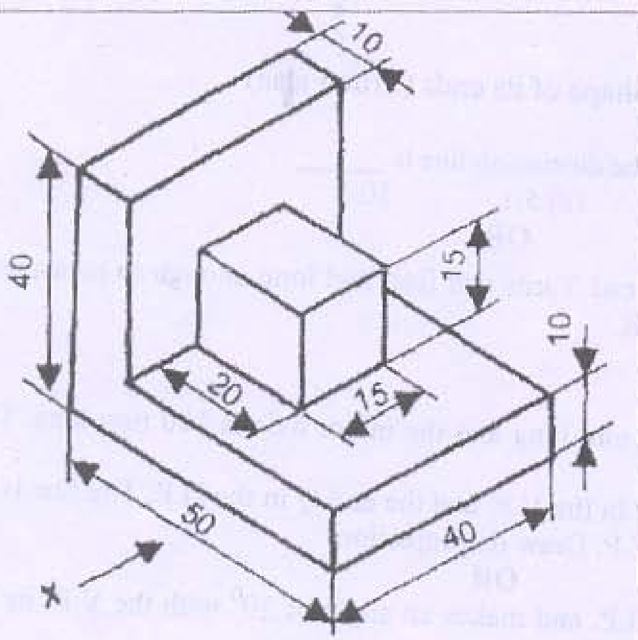

FIG.1

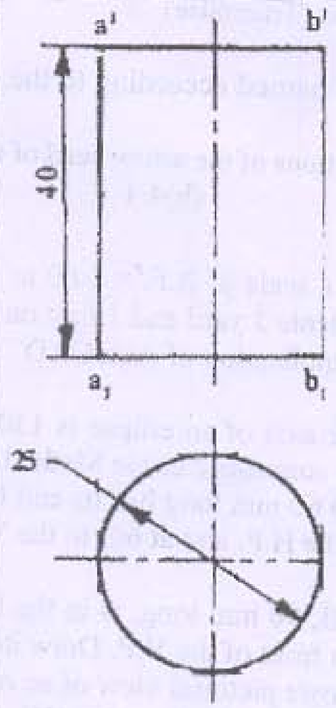

Fig-2